

Yuexuan Xu, MPP, PhD

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The Gertrude H. Sergievsky Center · Taub Institute for Research on Alzheimer's Disease and the Aging Brain
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RESEARCH OVERVIEW

I am a computational and genetic epidemiologist whose research focuses on elucidating the genetic and molecular mechanisms underlying Alzheimer's disease, with an emphasis on disease heterogeneity, molecular subtyping, and genomic and multi-omic risk prediction across diverse populations. My doctoral training emphasized genetic epidemiology, quantitative genetics, and gene-environment interactions, while my postdoctoral training has broadened this foundation through multi-ancestry genomics, biomedical data science, neurology, and multimodal biomarker research. I integrate large-scale genetic and genomic epidemiology studies with transcriptomic, metabolomic, proteomic, and epigenomic data, together with fluid, imaging, and neuropathological biomarkers, to characterize biologically distinct disease pathways and improve early detection and risk stratification. My work also applies advanced statistical and machine learning approaches to identify disease subtypes and model longitudinal cognitive and disease trajectories. The broader goal of my research is to advance precision medicine for Alzheimer's disease and related dementias by improving the identification, prediction, and biological characterization of heterogeneous disease processes. My independent research program will build on this foundation to define biologically distinct forms of dementia, identify molecular and environmental modifiers of genetic susceptibility, and understand how genetic risk translates into heterogeneity in pathology, disease progression, and clinical expression. My work has also received coverage from Newsweek, Medical Xpress, News-Medical, and Columbia University, reflecting broader clinical and public interest in this research.

ACADEMIC POSITIONS

2024–Present	Postdoctoral Research Scientist (Mentor: Dr. Richard Mayeux) The Gertrude H. Sergievsky Center & Taub Institute for Alzheimer's Disease and Aging Brain Columbia University (New York, NY, USA)
2024–Present	Honorary Associate Department of Population Health Sciences University of Wisconsin-Madison (Madison, WI, USA)
2020–2023	Research Assistant (Mentor: Dr. Corinne Engelman) Department of Population Health Sciences University of Wisconsin-Madison (Madison, WI, USA)
2019–2020	Research Assistant (Mentor: Dr. Jason Fletcher) Center for Demography of Health and Aging University of Wisconsin-Madison (Madison, WI, USA)
2018–2019	Research Specialist (PI: Dr. Titus Galama) Center for Economic and Social Research University of Southern California (Los Angeles, CA, USA)

EDUCATION

2019–2023	Doctor of Philosophy (Ph.D.) (<i>Degree conferred December 2023</i>) Emphasis: Statistics, Quantitative Genetics, and Genetic Epidemiology University of Wisconsin-Madison (Madison, WI, USA) Advisor: Dr. Corinne Engelman; Committee: Dr. Jason Fletcher, Dr. Lauren Schmitz, Dr. Qiongshi Lu
2016–2018	Master of Public Policy (M.P.P.) (<i>Degree conferred December 2018</i>) Emphasis: Social Science Genomics University of Southern California (Los Angeles, CA, USA)
2011–2015	Bachelor of Management (<i>Degree conferred July 2015</i>) Major: Public Administration; Emphasis: Social Statistics and Evaluation Research Shandong University (Jinan, Shandong, China)

PUBLICATIONS

Manuscripts in Preparation

- 1. **Xu Y**, et al. Pathology-Informed Polygenic Profiles Reveal Dementia Heterogeneity, Define Subtypes, and Improve Risk Prediction Across Ancestries. *Manuscript in preparation*.
- 2. **Xu Y**, et al. Longitudinal plasma phosphorylated tau 217 trajectories and risk of cognitive impairment across diverse populations. *Manuscript in preparation*.

Preprints

- 1. **Xu Y**, Sun Z, Jonaitis E, Deming Y, Lu Q, Johnson SC, Engelman CD. Mid-to-Late Life Healthy Lifestyle Modifies Genetic Risk for Longitudinal Cognitive Aging among Asymptomatic Individuals. *medRxiv* [Preprint]. 2024 May 27:2024.05.26.24307953. doi: 10.1101/2024.05.26.24307953. PMID: 38853902; PMCID: PMC11160812.

Peer-Reviewed Articles

- 1. **Xu Y**, Gunasekaran TI, Gu Y, Reyes-Dumeyer D, Piriz A, Sanchez D, Mejia DR, Medrano M, Lantigua RA, Alzheimer’s Disease Neuroimaging Initiative, Honig L, Wilson R, Reyes RER, Manly JJ, Brickman AM, Engelman CD, Johnson S, Asthana S, Bennett DA, Petersen M, O’Bryant S, Vardarajan BN, Mayeux R. Plasma phosphorylated tau 217 concentrations, APOE genotype, and timing of cognitive impairment in individuals across diverse racial and ethnic groups: a pooled analysis of prospective cohort studies. *Lancet Neurol*. 2026 Sep 9. doi: 10.1016/S1474-4422(26)00313-3. [Epub ahead of print] PMID: 42716078.
- 2. **Xu Y**, Qiao M, Gunasekaran TI, Gu Y, Reyes-Dumeyer D, Piriz A, Sanchez D, Soriano B, Franco Y, Coronado ZD, Recio P, Mejia DR, Medrano M, Lantigua RA, Honig L, Wilson R, Manly JJ, Brickman AM, Engelman CD, Johnson S, Asthana S, Vardarajan B, Mayeux R. Polygenic scores of Core 1 Alzheimer’s disease biomarkers predict early cognitive and pathological change. *Alzheimers Dement*. 2025 Nov;21(11):e70937. doi: 10.1002/alz.70937. PMID: 41307156; PMCID: PMC12658386.
- 3. **Xu Y**, Qiao M, Gunasekaran TI, Gu Y, Reyes-Dumeyer D, Piriz A, Sanchez D, Soriano B, Franco Y, Coronado ZD, Recio P, Mejia DR, Medrano M, Lantigua RA, Honig L, Manly JJ, Brickman AM, Vardarajan BN, Mayeux R. Evaluating polygenic risk score prediction performance for Alzheimer’s disease in a population-based Hispanic cohort using single- and multi-ancestry models. *Lancet Reg Health Am*. 2025 Jul 25;49:101198. doi: 10.1016/j.lana.2025.101198. PMID: 40747112; PMCID: PMC12312067.
- 4. **Xu Y**, Sun Z, Jonaitis E, Deming Y, Lu Q, Johnson SC, Engelman CD. Apolipoprotein E moderates the association between non-APOE polygenic risk score for Alzheimer’s disease and aging on preclinical cognitive function. *Alzheimers Dement*. 2024 Feb;20(2):1063–1075. doi: 10.1002/alz.13515. Epub 2023 Oct 19. PMID: 37858606; PMCID: PMC10916952.
- 5. **Xu Y**, Vasiljevic E, Deming YK, Jonaitis EM, Kosciak RL, Van Hulle CA, Lu Q, Carboni M, Kollmorgen G, Wild N, Carlsson CM, Johnson SC, Zetterberg H, Blennow K, Engelman CD. Effect of Pathway-Specific Polygenic Risk Scores for Alzheimer’s Disease (AD) on Rate of Change in Cognitive Function and AD-Related Biomarkers Among Asymptomatic Individuals. *J Alzheimers Dis*. 2023;94(4):1587–1605. doi: 10.3233/JAD-230097. PMID: 37482996; PMCID: PMC10468904.

RESEARCH SUPPORT & GRANTS

2027–2032	Principal Investigator <i>Clinical- and Biomarker-Informed Multi-Ancestry, Multi-Omic Prediction and Molecular Subtyping of Alzheimer’s Disease</i> NIH/NIA K99/R00 Pathway to Independence Award Status: Pending Role: PI Total costs: \$1,011,600.00
2022–2023	Project Lead (Faculty PI: Corinne Engelman) <i>Gene-Environment Interactions in Cognitive Decline During Preclinical Alzheimer’s</i> UW-Madison CDHA pilot project award, NIH P30AG017266 Status: Funded Role: Conceived the project, wrote the funded proposal, and led the research as my dissertation project Total costs: \$50,000

SEMINAR (INVITED) TALKS

- 2023 *"Beyond the Surface: Genes, Environment, and Their Interaction in Preclinical Alzheimer's Disease"*
Department of Psychiatry and Behavioral Sciences, University of California, San Francisco
Department of Population Health Sciences and Epidemiology, Stanford University
Department of Genetics and Genomic Sciences, Icahn School of Medicine at Mount Sinai

CONFERENCE PRESENTATIONS

Peer-Reviewed Abstracts

- 2026 *"Pathology-Informed Polygenic Scores Identify Distinct Genetic Contributors to Alzheimer's and Non-Alzheimer's Dementia Across Ancestries."* Accepted for poster presentation at the Clinical Trials on Alzheimer's Disease Conference 2026, Boston, MA, USA.
- "Plasma P-tau217 and APOE Genotype Can Predict Onset of Cognitive Impairment Before the Diagnosis of Alzheimer's Disease."* Accepted for oral presentation at the Clinical Trials on Alzheimer's Disease Conference 2026, Boston, MA, USA.
- "Pathology-Informed Polygenic Scores Identify Distinct Genetic Contributors to Alzheimer's and Non-Alzheimer's Dementia Across Ancestries."* Accepted for poster presentation at the Alzheimer's Association International Conference 2026, London, UK.
- "Plasma P-tau217 and APOE Genotype Can Predict Onset of Cognitive Impairment Before the Diagnosis of Alzheimer's Disease."* Accepted for oral presentation at the AD/PD Conference 2026, Copenhagen, Denmark.
- 2025 *"Polygenic scores of Core 1 Alzheimer's disease biomarkers predict early cognitive and pathological change."* Accepted for poster presentation at the Alzheimer's Association International Conference 2025, Toronto, Canada.
- "Polygenic scores of Core 1 Alzheimer's disease biomarkers predict early cognitive and pathological change."* Accepted for poster presentation at the AD/PD Conference 2025, Vienna, Austria.
- 2024 *"Evaluating polygenic risk score prediction performance for Alzheimer's disease in a population-based Hispanic cohort using single- and multi-ancestry models."* Accepted for poster presentation at the American Society for Human Genetics Annual Conference 2024, Denver, CO, USA.
- "Sex Differences in the Risk of Alzheimer's Disease and Associated Risk Factors Among Aging Hispanics: A Cross-country Study."* Accepted for poster presentation at the Alzheimer's Association International Conference 2024, Philadelphia, PA, USA.
- "Non-APOE Polygenic risk score derived from European ancestry data are more predictive of dementia among Hispanics who are APOE ϵ 4 carriers."* Accepted for oral presentation at the Alzheimer's Association International Conference 2024, Philadelphia, PA, USA.
- 2023 *"Childhood self-reported socioeconomic status modifies the genetic risk of Alzheimer's on mid- and late-life cognitive decline in a population-based study."* Accepted for poster presentation at the Alzheimer's Association International Conference 2023, Amsterdam, Netherlands.
- "Childhood self-reported socioeconomic status modifies the genetic risk of Alzheimer's on mid- and late-life cognitive decline in a population-based study."* Accepted for poster presentation at the Society for Epidemiology Annual Conference 2023, Portland, OR, USA.
- "Mid-to-Late Life Healthy Lifestyle Modifies Genetic Risk for Longitudinal Cognitive Aging among Asymptomatic Individuals from the Wisconsin Registry for Alzheimer's Prevention."* Accepted for poster presentation at the Wisconsin Alzheimer's Disease Research Day 2023, Madison, WI, USA.

2022	<i>"Mid-to-Late Life Healthy Lifestyle Modifies Genetic Risk for Longitudinal Cognitive Aging among Asymptomatic Individuals from the Wisconsin Registry for Alzheimer's Prevention."</i> Accepted for oral presentation at the Synergies for Longitudinal Study Conference, Madison, WI, USA. <i>"Mid-to-Late Life Healthy Lifestyle Modifies Genetic Risk for Longitudinal Cognitive Aging among Asymptomatic Individuals from the Wisconsin Registry for Alzheimer's Prevention."</i> Accepted for oral presentation at the Alzheimer's Association International Conference 2022, San Diego, CA, USA. <i>"Apolipoprotein E moderates the association between Non-APOE Polygenic Risk Score for Alzheimer's Disease and Aging on Preclinical Cognitive Function."</i> Accepted for poster presentation at the Alzheimer's Association International Conference 2022, San Diego, CA, USA. <i>"Apolipoprotein E moderates the association between Non-APOE Polygenic Risk Score for Alzheimer's Disease and Aging on Preclinical Cognitive Function."</i> Accepted for poster presentation at the Wisconsin Alzheimer's Disease Research Day 2022, Madison, WI, USA.
2021	<i>"Effect of Pathway-specific Polygenic Risk Scores for Alzheimer's Disease on Rate of Change in Cognitive Function and AD-related Biomarkers among Asymptomatic Individuals."</i> Accepted for oral presentation at the Integrating Genetics into the Social Sciences Conference 2021, Boulder, CO, USA.
2020	<i>"Bald is Beautiful? Life course outcomes for individuals at high genetic risk for male pattern baldness."</i> Accepted for oral presentation at the Integrating Genetics into the Social Sciences Conference 2020, Boulder, CO, USA.

TEACHING & MENTORING

2018	Teaching Assistant (Instructor: Dr. John Romley) Sol Price School of Public Policy University of Southern California (Los Angeles, CA, USA)
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HONORS/AWARDS

2021, 2023	<i>Graduate Student Research Competition Grant</i> University of Wisconsin-Madison
2022	<i>Rankin / Skatrud Research Award</i> University of Wisconsin-Madison
2019	<i>Chancellor's Fellowship</i> University of Wisconsin-Madison
2019	<i>PHS Outstanding Student Scholarship</i> University of Wisconsin-Madison
2018	<i>Phi Alpha Alpha National Honor Society</i> (inducted in 2018) Network of Schools of Public Policy, Affairs, and Administration
2015	<i>Outstanding Undergraduate Thesis Award</i> Shandong University
2013	<i>Scholarship for Academic Excellence</i> Shandong University

PROFESSIONAL SERVICE

Ad Hoc Reviewer	<i>Journal of Alzheimer's Disease; JAD Reports; Brain, Behavior, and Immunity – Health; Scientific Reports; International Journal of Geriatric Psychiatry; Dementia and Geriatric Cognitive Disorders; International Journal of Molecular Sciences</i>
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MEDIA COVERAGE & RESEARCH FEATURES

2026	U.S. National Institutes of Health (NIH) — Official U.S. Government Research Feature (NIH Research Matters) — <i>Estimating Cognitive Impairment Risk, Timing</i> October 2026 (https://www.nih.gov/news-events/nih-research-matters/estimating-cognitive-impairment-risk-timing) Newsweek — <i>Alzheimer's Test Could Reveal When Symptoms May Begin Years in Advance</i> September 2026 (https://www.newsweek.com/alzheimers-test-could-reveal-when-symptoms-may-begin-years-in-advance-12447970)
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International Business Times Australia — *New Blood and Gene Test Combo Can Predict Alzheimer's Symptoms Years Before They Appear, Study Finds* | September 2026 (<https://www.ibtimes.com.au/blood-test-apoe4-predict-alzheimers-symptoms-1875571>)

News-Medical — *APOE Genotype Helps p-tau217 Predict Who May Develop Cognitive Impairment Sooner* | September 2026 (<https://www.news-medical.net/news/20260914/APOE-genotype-helps-p-tau217-predict-who-may-develop-cognitive-impairment-sooner.aspx>)

Commonwealth Union — *Can a Blood Test Predict Alzheimer's Before Memory Loss Begins? New Study Offers Clues* | September 2026 (<https://www.commonwealthunion.com/can-a-blood-test-predict-alzheimers-before-memory-loss-begins-new-study-offers-clues/>)

ALLATRA Media — *A Gene Result Changes What an Alzheimer's Blood Test Can Tell You* | September 2026 (<https://allatra.media/news/science-ptau217-apoe-timing-of-cognitive-impairment>)

Medical Xpress — *Blood and Gene Tests Predict Alzheimer's Symptoms Three to Six Years Ahead* | September 2026 (<https://medicalxpress.com/news/2026-09-blood-gene-alzheimer-symptoms-years.html>)

Columbia University Irving Medical Center — *Blood Biomarker and Genetic Test May Help Predict Timing of Cognitive Impairment* | September 2026 (<https://www.neurology.columbia.edu/news/blood-biomarker-and-genetic-test-may-help-predict-timing-cognitive-impairment>)

Science — *An Alzheimer's Blood Test Is Getting Popular. But Who Really Needs It?* | March 2026 (<https://www.science.org/content/article/alzheimer-s-blood-test-getting-popular-who-really-needs-it>)

2025

TaubCONNECT Research Perspectives: October 2025

<https://columbianeuroresearch.org/taub/res-taub-connect-10-2025.html#4>

TaubCONNECT Research Perspectives: December 2025

<https://columbianeuroresearch.org/taub/res-taub-connect.html>

EXTERNAL TRAINING, CERTIFICATES, AND WORKSHOPS

2022	<i>Novel models for longitudinal continuous outcomes workshop</i> Stanford University
2021	<i>DNA methylation analysis workshop</i> University of Colorado-Boulder
2020	<i>Statistical workshop on genomic family data</i> University of Colorado-Boulder
2020	<i>Genome-wide data analyses</i> Tinbergen Institute and University of Wisconsin-Madison

TECHNICAL EXPERTISE & ADDITIONAL INFORMATION

eRA Commons:	YUEXUANXU
Google Scholar:	scholar.google.com/citations?user=dfyaxs8AAAAJ
Languages:	English; Chinese
Software:	R, SAS, Stata, ArcGIS, PLINK/PLINK2, GCTA, LDSC, PC-AiR, PC-Relate, PRSice-2, PRS-CS, LDpred2, METAL, REGENIE, SAIGE, KING, ADMIXTURE, EIGENSOFT, bcftools, VCFtools, BEDTools, Minimac4, Eagle, ANNOVAR, VEP, FUMA, MAGMA, PrediXcan, FUSION, DESeq2, limma, WGCNA, Cytoscape
Programming languages:	R (advanced), Python (proficient), Bash/Shell (proficient), SQL (working proficiency)
Computational environment:	High-performance computing (HPC) on Linux-based clusters (Slurm) and cloud computing environments (AWS) for large-scale genomic, multi-omic, and biomarker analyses
NCBI Bibliography:	ncbi.nlm.nih.gov/myncbi/yuexuan.xu.3/bibliography/public/

PROFESSIONAL EXPERIENCE

2017–2018	<i>Part-time policy analyst</i> (Master's Practicum / Capstone Project) International Refugee Assistance Project
Summer 2017	<i>Policy analyst intern</i> Fudan University PPIR Center
2015–2016	<i>Executive Intern, Office of the CEO</i> Shanghai Hanyu Environmental Protection Technology Co., Ltd.

PROFESSIONAL MEMBERSHIPS

International Society to Advance Alzheimer's Research and Treatment; American Society of Human Genetics; Society for Epidemiologic Studies; International Genetic Epidemiology Society

REFERENCES

Richard Mayeux, MS, MD

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Corinne D. Engelman, MSPH, PhD

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